PHILOSOPHY OF SCIENCE

LESSON 11
SCIENTIFIC PROCESSES AND OBJECTIVITY

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SUMMING UP LESSON 10

Confirmation bias:

Both about how we collect / process / obtain information

Jigsaw puzzles analogy →

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LESSON 11: AGENDA

- **Aschwanden 2015**
    - Science is difficult I – politics example
    - Is science broken due to cheating & biases?
    - Science is difficult II – soccer example
    - Is there a right scientific analysis of a study?
    - Science as a collective process & what kind of truth?

- **Harrer 2021**
    - What is a meta-analysis
    - Strenghts and pitfalls

Lesson 7-9
The principles of why it is difficult to reach true objectivity

Lesson 11
Practice-oriented look at the objectivity challenge and options to solve it

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ASCHWANDEN 2015: IS SCIENCE BROKEN?

Science troubled by

- Replication crisis
- Fraudelent papers getting published
- Researchers have biases
- Inconsistent results
- ....

Aim of todays lesson →

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MANY ORIGINAL STUDIES DO NOT REPLICATE? (WHEN REDONE BY OTHERS)

http://theneuroeconomist.com/2016/03/is-the-replication-crisis-in-psychology-real/

https://science.sciencemag.org/content/349/6251/aac4716

WHY THIS LACK OF REPLICATION?

Shouldn't science have institutions, peer review processes and criteria that ensures that only true / good research gets published?

PEER REVIEW CAN'T FIX EVERYTHING: CHEATING, (BAD?) INCENTIVES AND HUMAN BIASES

Partially due to fraud / cheating (see lesson 12)

- Some papers get retracted (0.02%) most due to fraud

Partially due to human biases (cf. Kahneman)

- "failling prey to human biases…"
- "you believe your hypothesis…"
- "you have to believe…to give energy and passion"

(all Aschwanden 2015 quotes)

https://twitter.com/lakens/status/133522764 5135564800?s=20

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P-(VALUE) HACKING

"That's because answering **even a simple scientific question** — which party is correlated with economic success — **requires lots of choices that can shape the results**." (Aschwanden 2015)

"The variables in the data sets you used to test your hypothesis had 1,800 possible combinations. Of these, 1,078 yielded a publishable p-value,[1] but that doesn't mean they showed that which party was in office had a strong effect on the economy. Most of them didn't." (Aswchwanden 2015, https://fivethirtyeight.com/features/science-isnt-broken/#part1)

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CHARLES BABBAGE, IN 1830: REFLECTIONS ON THE DECLINE OF SCIENCE IN ENGLAND

On the problem of bias in science, and lack of replications in science

An old problem!

https://twitter.com/lakens/status/1305153073258561542?s=20

ASCHWANDEN 2015: SCIENCE IS NOT BROKEN

Yet also better reasons for why science studies do not always replicate:

"...I've spent months investigating the problems hounding science, and I've learned that the headline-grabbing cases of misconduct and fraud are mere distractions. The state of our science is strong, but it's plagued by a universal problem: Science is hard — really fucking hard."

https://fivethirtyeight.com/features/science-isnt-broken/#part1

HOWEVER: WHY SAME DATA, DIFFERENT CONCLUSIONS?

IS THE CONCLUSION TRUE? ARE DARK-SKINNED PLAYERS MORE LIKELY TO GET A RED CARD? (VEVOX.APP 143-136-950)

- Yes, true
- No, not all studies show this result. Maybe they were (politically, cognitively, emotionally) biased?
- Well, what does 'truth' mean?

NO BIAS? WELL, NO PRIOR SYSTEMATIC DIFFERENCES BETWEEN THE 29 TEAMS

For example
Research team's prior (personal) beliefs about an expected relation unrelated to result

Of course difficult to completely rule out impact of any prior bias

Twenty-nine research teams were given the same set of soccer data and asked to determine if referees are more likely to give red cards to dark-skinned players. Each team used a different

Table 4. Covariates Included by Each Team

Nelson 2016: Heterogeneous constructs and numerous influences…

Is there always a correct decision to be made?

DOI: (10.1177/2515245917747646)

Many decisions

Fig. 3. Point estimates (clustered by analytic approach) and 95% confidence intervals for the effect of soccer players' skin tone on the number of red cards awarded by referees. Reported results, along with the analytic approach taken, are shown for each of the 29 analytic teams. The teams are clustered according to the distribution used in their analyses; within each cluster, the teams are listed in order of the magnitude of the reported effect size, from smallest at the top to largest at the bottom. The asterisks indicate upper bounds that have been truncated to increase the interpretability of the plot (see Fig. 2). OLS = ordinary least squares; WLS = weighted least squares; Misc = miscellaneous.

DOI: (10.1177/2515245917747646)

IS THERE EVEN SUCH A THING AS THE RIGHT ANALYSIS?
WELL, SOME CHOICES ARE BETTER THAN OTHERS

Some analysis is wrong

Maybe multiple analyses can be right?

https://twitter.com/lakens/status/1166049199470850049?s=20

DIFFERENT INTERPRETATION OF RESEARCH QUESTION → DIFFERENT CHOICES

”In 2018...published an article in which 29 teams analyzed the same research question with the same data: ... In this article, we investigate why results diverged so much. **We argue that the main reason was an unclear research question: Teams differed in their interpretation of the research question and therefore used diverse research designs and model specifications. We show by reanalyzing the data that with a clear research question, a precise definition of the parameter of interest and theory-guided causal reasoning, results vary only within a narrow range**. The broad conclusion of our reanalysis is that social science research needs to be more precise in its “estimands” to become credible.”

https://doi.org/10.1177/23780231211024421

Less variation in results!

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ANOTHER EXAMPLE: MANY ANALYSTS, MANY (QUITE?) DIFFERENT RESULTS

- 162 researchers (73 teams).
- Data: 6 questions from the International Social Survey Programme
- Hypothesis: (whether) immigration reduces public support for social policies
- Results: substantial variation, not predicted by researcher skills or beliefs

- Some (not all!) questions are just really difficult – many choices to be made

https://twitter.com/tomstafford/status/1377131265405505546

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SAME (RED CARD) DATA, MOSTLY SIMILAR CONCLUSIONS

- "On the other hand, **it also suggests there's a *there* there.** It's hard to look at that data and say there's no bias against dark-skinned players."

- "The important lesson here is that a single analysis is not sufficient to find a definitive answer. **Every result is a temporary truth,** one that's subject to change when someone else comes along to build, test and analyze anew."

- "But these disparate results don't mean that studies can't **inch us toward truth**."

(Aschwanden 2015)

http://www.proprofs.com/quiz-school/user_upload/ckeditor/Trooth.jpg

SCIENTIFIC DEVELOPMENT: IN PRACTICE

"The standard way of thinking about the scientific method is: ask a question, do a study, get an answer. But this notion is vastly oversimplified. A more common path to truth looks like this: ask a question, do a study, get a partial or ambiguous answer, then do another study, and then do another to keep testing potential hypotheses and homing in on a more complete answer. Human fallibilities send the scientific process hurtling in fits, starts and misdirections instead of in a straight line from question to truth."

(Aschwanden 2015)

http://undsci.berkeley.edu/images/us101/paths.gif

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SCIENTIFIC DEVELOPMENT: COLLECTIVE PROCESS

"Since people are fallible, and often obstinate and overly fond of their own ideas, the objectivity of the process which tests conjectures lies not in the emotional detachment and impartiality of individual scientists, but rather in the scientific community being organized in certain ways, with certain institutions, norms and traditions, so that individuals' prejudices more or less wash out (Popper, 1945, Chapters 23–24)."

http://www.stat.columbia.edu/~gelman/research/published/philosophy.pdf

≈ Kahneman, Kuhn

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KUHN VS ASCHWANDEN: WHAT DOES THE WORD 'TRUTH' ACTUALLY MEAN?

But the scientists have not realized that Kuhn's claim is based on **a philosopher's definition of "truth," which is not achieved** (if ever) until there is exact knowledge of the ultimate constituents of matter on the quantum-gravity scale. **Scientists use a less demanding definition for the word "truth**," in which measured parameters can be subject to nonzero error bars."
Kenneth G. Wilson, in *Physics Today* 54, 3, 53 (2001)

Kuhn

Aschwanden

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A RELATED METAPHOR, ON TRUTH

"The empirical basis of science has nothing 'absolute' about it. Science does not rest upon solid bedrock. The bold structure of its theories rise, as it were above a swamp. It is like a building erected on piles. The piles are driven down from above into the swamp, but not down to any natural or 'given' base; and if we stop driving the piles deeper, it is not because we have reached firm ground. We simply stop when we are satisifed that the piles are firm enough to carry the structure at least for the time being."
(Popper 1959, p. 111 quoted from Duberley & Johnnson 2000,)

SCIENCE AS UNCERTAINTY REDUCTION I

WE OFTEN SHOULDN'T WAIT FOR 100% CERTAINTY

"Science is not a magic wand that turns everything it touches to truth. Instead, "science operates as a procedure of uncertainty reduction," said Nosek, of the Center for Open Science. "The goal is to get less wrong over time." (Aschwanden 2015)

https://twitter.com/erikleejohnson/status/1357413415606755334?s=20

Reflect 2-3 min: What is Aschwandens (2015) primary message?

"Science is not a magic wand that turns everything it touches to truth. Instead, "science operates as a procedure of uncertainty reduction," said Nosek, of the Center for Open Science. "The goal is to get less wrong over time." (Aschwanden 2015)

SO, WHY IS SCIENCE NOT BROKEN?
(OR WHY SHOULD THE PUBLIC BELIEVE IN SCIENCE)

Yes, researchers have biases, some cheat, some are sloppy. But →

- Science is ******* difficult, failures are to be expected
- Some fields more heterogeneous, fuzzy and difficult than others
- Researchers are fallible and biased, paradigmatic groupthink constrains
- Don't trust one study, trust multiple studies with different research designs
- Be critical: What is measured (and where), what is theorized?

Thus: Avoid these 2 flawed interpretations of science

- **Pure relativism → because we collectively can't just produce any result**
- **Blind belief in science → because we collectively don't just produce one result**

This video by John Oliver relates to today's issue: https://www.youtube.com/watch?v=0Rnq1NpHdmw

2021 SPRING EXAM QUESTION (DISCUSS 2-3 MIN)

Question 3 (20%)

i) Please identify a quote in Aschwanden (2015) that you think has the best fit with the overall message in Nelson (2016). Provide an explanation for how the quote fits into Nelson's overall message. ii) Please also identify a quote in Aschwanden (2015) that seems to be in disagreement with Nelson (2016).

Note that a quote can be one or several sentences long.

I will upload a good exam answer to Brightspace

ASCHWANDEN AND NELSON: 2021 GRADING GUIDE

Several quotes can be applied in i). The following lists some of the most obvious ones. Again, others can be applied. Arguments should be presented, both on Nelson's position (not elaborated here), but also on how the quote links to Nelson.

- "*That's because answering even a simple scientific question — which party is correlated with economic success — requires lots of choices that can shape the results. This doesn't mean that science is unreliable. It just means that it's more challenging than we sometimes give it credit for.*" - this is probably the most ideal quote
- "*Even the most skilled researchers must make subjective choices that have a huge impact on the result they find*"
- "*If we are going to rely on science as a means for reaching the truth – and it's still the best tool we have – it's important that we understand and respect just how difficult it is to get a rigorous result.*"
- "*As a society, our stories about how science works are also prone to error. The standard way of thinking about the scientific method is: ask a question, do a study, get an answer. But this notion is vastly oversimplified*"
- "*The important lesson here is that a single analysis is not sufficient to find a definitive answer.*"
- "*The uncertainty inherent in science doesn't mean that we can't use it to make important policies or decisions. It just means that we should remain cautious and adopt a mindset that's open to changing course if new data arises.*"

Some rely on the following quote: "The state of our science is strong but it's plagued by a universal problem: Science is hard - really fucking hard." Nelson would not disagree as such, but since this quote is not specific, using this quote requires one to specify what exactly Aschwanden means, in order to enable a direct comparison to Nelson's position.

A relevant quote for ii)

- "*"But these disparate results don't mean that studies can't inch us toward truth."* Several other similar quotes can be applied. The position one can unfold is that Aschwanden's position is more akin to post-positivism, while Nelson's position is more akin to constructivism..
- "*P-hacking and similar types of manipulations often arise from human biases.*" - one could use this quote to state that while Aschwanden focuses somewhat on biases and p-hacking, Nelson argues that social science is difficult because of the nature of the subject, rather than the biased humans studying social science.
- Other quotes are also useful.

LINKING ASCHWANDEN TO PRIOR CURRICULUM

Lesson 10 (Kahneman): Human biases shape decisions

Lesson 8 (Nelson): Heterogeneous variables, numerous influences

Lesson 7-8 (Kuhn): Paradigms and prior beliefs / standards in scientific communities

Lesson 9 (Guba): Are the effects real and true?

Lesson 2 (Isager): Difficult to identify relevant confounding variables

Lesson 6 (Theory): Induction vs. deduction (inductive search or ONE test?)

FROM ASCHWANDEN TO HARRER ET AL. 2021: META-ANALYSES

How to improve objectivity?

- Aschwanden: one dataset, many analyses
- Meta-analysis: many datasets, one analysis

https://www.stevestewartwilliams.com/p/exercise-can-be-more-effective-than

HOW TO MAKE SENSE OF A DIVERSE RESEARCH FIELD?

What is the answer to X question? Go to literature and check

But 1) there is a lot of research, and 2) (cf. Aschwanden 2015) not all prior research is true?

Need to ”critically appraise bodies of evidence **in their entirety**” (Harrer et al. 2021, p. 2)

HOW TO DIGEST AN ENTIRE FIELD? (P. 2-3)
AND WHAT IS A META-ANALYSIS

How to summarize the evidence on a given topic? (e.g. if exercise → reduces depression)

- Traditional / narrative reviews: No strict rules, risk of bias – qualitative conclusions
- Systematic reviews: Summarizes evidence based on pre-defined and transparent standards – qualitative conclusions

- **Meta-analysis: Systematic selection and statistical analysis – quantitatively**
  - Averages effects across studies; e.g., Study 1: 0.3 effect, Study 2: 0 effect, Study 3: 0.5 effect → meta-analytic average: 0.27 effect
  - An analysis of existing analyses: Prior studies become units of analysis
  - Only relevant for quantitative results

- Sounds like an easy solution? Well, still difficult

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HOW TO DO A META-ANALYSIS

Define what studies to include

- What kind of exercise (independent variable)
- What kind of mood measurements (dependent variable)
- What sample criteria: all populations, all kinds of study designs, all time periods?

Specify how analysis is to be done

- Weighted average of effect sizes (bigger samples / better studies more impact e.g.)
- Sensitivity analysis (i.e. does it matter exactly how analysis is done (cf. Aschwanden))

Should multiple meta-analyses on the same topic lead to identical results?

Well, pitfalls exist → next slides

BRIEF EXPLAINER OF META-ANALYSES

https://www.youtube.com/watch?v=i675gZNe3MY

COMPARING APPLES AND ORANGES? (P. 7) I

"One may argue that meta-analysis means combining apples with oranges. Even with the strictest inclusion criteria, studies in a meta-analysis will never be absolutely identical. There will always be smaller or larger differences between the included sample, the way an intervention was delivered, the study design, or the type of measurement used in the studies.

This can be problematic. Meta-analysis means to calculate a numerical estimate which represents the results of all studies. Such an estimate can always be calculated from a statistical point of view, but it becomes meaningless when studies do not share the properties that matter to answer a specific research question."

COMPARING APPLES AND ORANGES? III

If very consistent results, across interventions / variables → less to worry about (?)

GARBAGE IN, GARBAGE OUT

The quality of evidence produced by a meta-analysis heavily depends on the quality of the studies it summarizes. If the results reported in our included findings are biased, or downright incorrect, the results of the meta-analysis will be equally flawed. This is what the "Garbage In, Garbage Out" problem refers to. It can be mitigated to some extent by assessing the quality or **risk of bias** (see Chapter 1.4 and 15) of the included studies.
However, if many or most of the results are of suboptimal quality and likely biased, even the most rigorous meta-analysis will not be able to balance this out

Harrer et al. 2021, p. 8

"When we average valid with invalid studies, the resulting average is invalid" https://datacolada.org/104

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FILE DRAWER PROBLEM / PUBLICATION BIAS (P. 8)

"The file drawer problem refers to the issue that not all relevant research findings are published, and are therefore missing in our meta-analysis."

"There is good reason to believe that studies with negative or "disappointing" results are systematically underrepresented in the published literature and that there is a so called publication bias. The exact nature and extent of this bias can be at best a "known unknown" in meta-analyses." (Harrer et al. 2021, p. 8)

One can check how symmetric published results are: more results just barely significant, than just barely non-significant?

Scenario A
Researcher 1 does study on X impact on Y, finds no (interesting) result
Doesn't manage to get study published

Scenario B
Researcher 2 does study on X impact on Y, finds (interesting) result.
Manages to get study published

Scenario C....

Result: Published studies are biased

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Links to Kuhn's (constraining) paradigms

RESEARCHER BIAS?

“There is evidence that, given one and the same data set, even experienced

analysts with the best intentions can come to drastically varying conclusions

(Silberzahn et al. 2018 [the red card study]).” (Harrer et al. 2021, p. 9)

”One way to reduce the researcher agenda problem is pre-registration, and publishing a detailed analysis plan before beginning with the data collection for a meta-analysis” (Harrer et al. 2021, p. 9)

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WHAT IS THE BEST KIND OF STUDY / GOLD STANDARD?

What to do, if searching for scientific answer to a question?

- Search literature, take research designs into consideration
- Preferably find a systematic review
- Ideally a meta-analysis
  - Still not bullet-proof cf. pitfalls)
  - Different meta-analyses on same topic can provide different results

A post-positivistic approach

Focused on quantitative challenges

Modified from: https://www.researchgate.net/publication/336191285/figure/fig3/AS:843719670505473@1578169853483/ASPAN-hierarchy-of-evidence-ASPAN-American-Society-of-PeriAnesthesia-Nurses-O-Laura.jpg

SUMMARY LESSON 11

- Science is complicated and produced by biased researchers that sometimes cheat
- Not all science results replicate → non-trustworthy science? Can we make anything up?

- Aschwanden: How one dataset can be analyzed in many ways → if most agree, probably inching towards truth
    - From individual subjectivity to collective intersubjectivity / 'objectivity
    - While multiple choices might be right, many will be wrong
    - Trust many studies, not one – yet, how to interpret many studies?

- Harrer: From narrative reviews to quantitative meta-analysises → if consistent results, probably uncovered something that is indicative?
    - Risk of comparing apples and oranges
    - Publication bias
    - Garbage in-garbage out
    - Researcher bias not eliminated

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OMITTED VARIABLE BIAS AND CONFOUNDING VARIABLES

Omitted Variable Bias occurs when a variable that should be included in the analysis is left out. This can result in a biased estimate of the relationship between the independent variable (IV) and the dependent variable (DV). If the omitted variable is correlated with the IV, this can lead to an over- or underestimation of the IV's effect on the DV, because the model incorrectly attributes the effect of the omitted variable to the IV.

For example, if we're investigating the effect of gender diversity (IV) on firm performance (DV), but we omit a variable like "managerial experience," which could influence firm performance, we may incorrectly attribute the effect of managerial experience on performance to gender diversity if managerial experience is also correlated with gender diversity.

Confounding Variable Bias occurs specifically when an omitted variable is related to both the IV and the DV, potentially providing an alternative explanation for an observed association. This confounder can falsely suggest that there is a direct cause-and-effect relationship between the IV and DV.

Using the same example, if we do not include a variable such as "industry growth" that is related to both gender diversity (IV) and firm performance (DV), any observed effect of gender diversity on firm performance might be confounded by the growth of the industry. That is, industries that are growing may both perform better and attract a more diverse workforce, leading to the incorrect conclusion that gender diversity causes better firm performance, when in fact industry growth is influencing both.

In summary, omitted variable bias can involve variables that are related to either the DV or the IV, causing biased estimates. Confounding variable bias is a specific type of omitted variable bias where the omitted variable has a relationship with both the DV and the IV, misleadingly suggesting a causal relationship between the two.

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